

Parallel and Perpendicular Lines

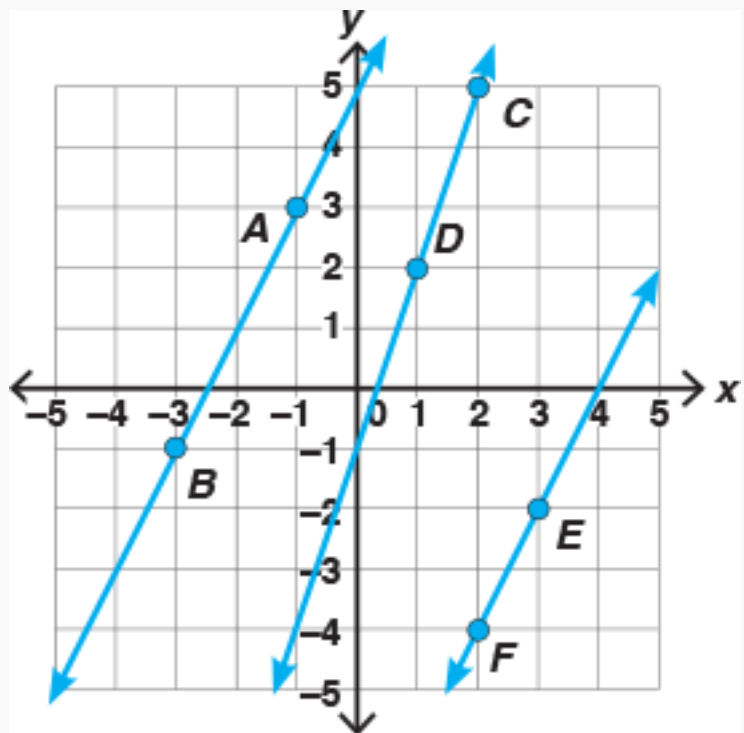
You have already learned the definitions of **parallel** and **perpendicular**. Parallel lines will never intersect. Perpendicular lines intersect at right angles. You can recognize these types of lines on a coordinate plane by looking at their slopes.

Parallel lines have the same slope.

Example

Hide

On the coordinate grid below, which two lines are parallel?



Find the slopes of all three lines.

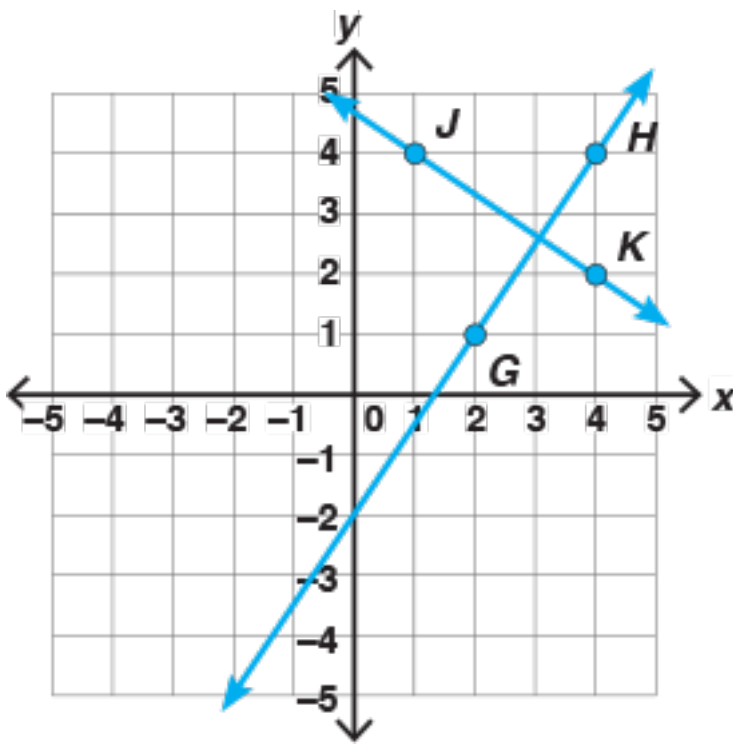
Line AB : $m = \frac{3 - (-1)}{-1 - (-3)} = \frac{4}{2} = 2$

Line CD : $m = \frac{5 - 2}{2 - 1} = \frac{3}{1} = 3$

Line EF : $m = \frac{-2 - (-4)}{3 - 2} = \frac{2}{1} = 2$

Line AB and **line EF** have the same slope. They are parallel.

Look at the perpendicular lines graphed on the coordinate grid. Can you see that they go in opposite directions?



A line that *rises* from left to right has a **positive slope**.

A line that *falls* from left to right has a **negative slope**.

Find the slopes of these two perpendicular lines.

Line GH : $m = \frac{4 - 1}{4 - 2} = \frac{3}{2}$ Line JK : $m = \frac{4 - 2}{1 - 4} = -\frac{2}{3}$

Because the lines go in opposite directions, one slope is positive and the other slope is negative. The fractions are also inverted. This type of opposite is called a **negative reciprocal**. $-\frac{2}{3}$ is the negative reciprocal of $\frac{3}{2}$.

For perpendicular lines, the slope of one line is the negative reciprocal of the slope of the other line.

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